

Support Vector machines:



$$\pi_2 > \pi_1?$$

as

$$z = \sum w_i x_i$$

$$\sigma(z)/p_1 = 0.1$$

(close)

$$\sigma(z)/p_2 = 0.9$$

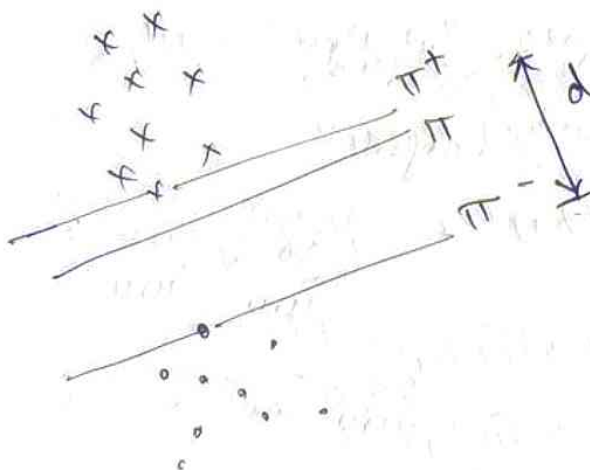
more is distance of line and pt,
more is $P(X)$

SVM: Minimise the distance $(w^T x_i)$ to ~~maximize~~ classify points as widely as possible.

$\pi_2 \rightarrow$ margin maximizing hyperplane.

Steps:

1. Draw a plane $\parallel$ to π^* , touching the ^{upper} closest point (π^+)
2. " " " " " " " lower point (π^-)
3. $d = |\pi^+ - \pi^-|$ (max value of d , best hyperplane)
 $\rightarrow$ margin.

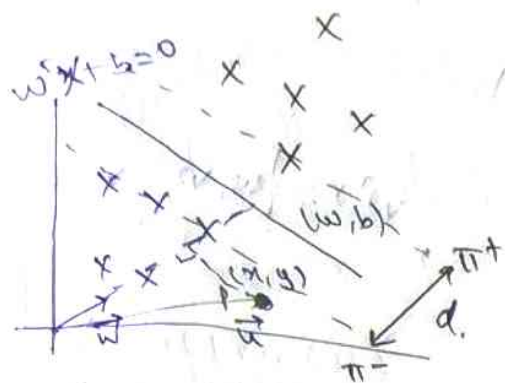


1) Robust to outliers

2) Non linear data

3) Can be implemented
in both Regress,
classification.

maths:



$\vec{w}$ is $\perp$ to π plane.
projection of $\vec{u}$ on $\vec{w}$

$$\vec{u} \cdot \hat{\vec{w}} = \frac{\vec{u} \cdot \vec{w}}{|\vec{w}|} \rightarrow 1 \text{ (unit)}$$

for $P(x|y)$ to be true (x)

$$\vec{u} \cdot \vec{w} \geq c$$

$$\vec{w} \cdot \vec{u} - c \geq 0$$

$$\vec{w} \cdot \vec{u} + b \geq 0$$

true.

point to be classified.

for any $\vec{x}_i$,

$$\hat{y} = \begin{cases} +ve (+1) & \vec{w} \cdot \vec{x}_i + b \geq 0 \\ -ve (-1) & \vec{w} \cdot \vec{x}_i + b < 0 \end{cases}$$

$$\text{Let } \pi^+ : w^T x + b = 1$$

$$\pi^- : w^T x + b = -1$$

q?

for a given hyperplane.

$$w_0 + \sum_{i=1}^n w_i x_i = 0$$

$$w_0 + (w_1 x_1 + w_2 x_2 \dots w_n x_n) = 0$$

$$\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} n \times 1$$

$$w_0 + [w_1 \ w_2 \ \dots \ w_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = 0$$

$$w_0 + w_1 x_1 + w_2 x_2 \dots = 0$$

$$\begin{bmatrix} w_0 & w_1 & w_2 & \dots & w_n \end{bmatrix} \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = 0$$

(1×1) $(n+1) \times 1$ (1×1)

~~$$W^T X = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}$$

$$n \times 1 \quad 1 \times n$$~~

$$W^T X = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$W^T X = [w_1, w_2, \dots, w_n]$$

$$1 \times n$$

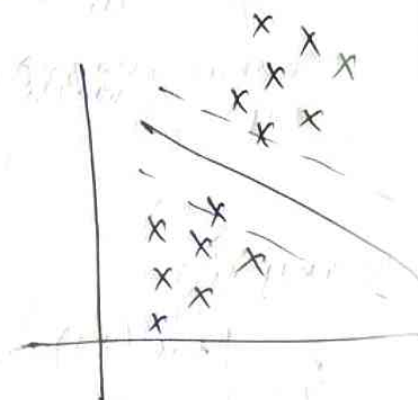
(1x1)

$$w_0 + W^T X = 0$$

$$W^T X + b = 0 \quad (\text{eqn of hyperplane})$$

for all (x) green points
to be above line,

$$\vec{w} \cdot \vec{x} + b > 1$$



for all red (x) points

$$\vec{w} \cdot \vec{x} + b \leq -1$$

for a single eqn, create a label

$$y_i = 1 \quad (\text{green})$$

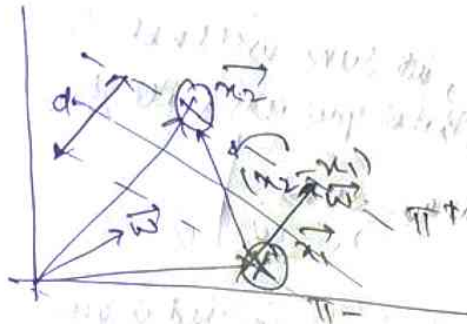
$$y_i = -1 \quad (\text{red})$$

$$\text{rule} \quad \begin{cases} y_i (\vec{w} \cdot \vec{x} + b) > 1 & \rightarrow (\text{green}) \\ y_i (\vec{w} \cdot \vec{x} + b) < -1 & \rightarrow (\text{red}) \end{cases}$$

hence,

$$y_i (\bar{w} \cdot \bar{x} + b) > 1$$

for constraint, (red points lying on π^+, π^-)
(support vectors)



for support vectors (x_2, x_1)

$$y_i (\bar{w} \cdot \bar{x} + b) = 1$$

for d , if we find projection of $(x_2 - x_1)$ on $\bar{w}$, it is d .

as b to π, π^+, π^-

convert $\bar{w}$ to unit,

$$\hat{w} = \frac{w}{\|w\|} \rightarrow \text{mag.}$$

$$d = \frac{(x_2 - x_1) \cdot \bar{w}}{\|w\|}$$

$$d = \frac{x_2 \cdot \bar{w} - x_1 \cdot \bar{w}}{\|w\|}$$

$$\text{for } x_2, (y_i)(w x_2 + b) = 1$$

$$w x_2 = 1 - b$$

$$w x_1$$

$$(y_i)(w x_1 + b) = 1$$

$$(-1)(w x_1 + b) = 1 \Rightarrow w x_1 = -1 - b$$

$$d = \frac{(1-b) - (-1-b)}{\|w\|} = \frac{1-b+1+b}{\|w\|}$$

$$d = \frac{2}{\|w\|}$$

argmax $\frac{2}{\|w\|}$ such that.

$$(w^*, b^*)$$

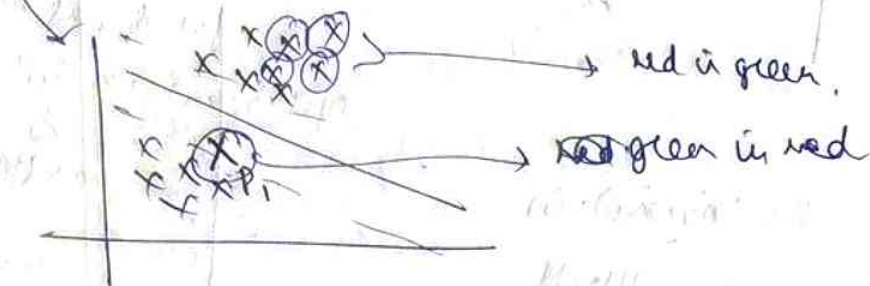
$$(y_i)(\bar{w} \bar{x} + b) \geq 1$$

(find a w, b that maximizes $\frac{2}{\|w\|}$)

* Problem:

if data is non linear, SVM will not find π^+, π^- and process will fail.

called Hard margin SVM



Soft margin SVM:

$$\text{for } P_i, (y_i)(\bar{w} \bar{x} + b) \geq 1.$$

$$y_i = +1 \text{ (As green point)}$$

$$\bar{w} \cdot \bar{x} + b \leq -1 \text{ (As in red region)}$$

$$(y_i)(\bar{w} \bar{x} + b) < -1$$

so Hard margin SVM fails.

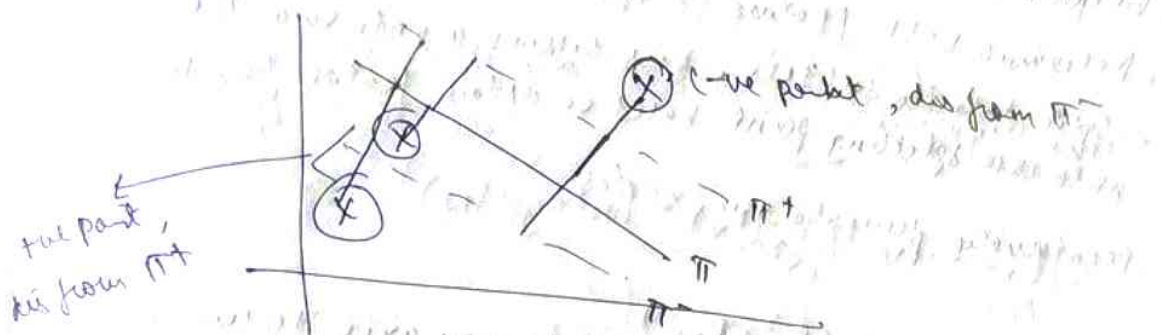
for soft margin

$$\text{argmax}_{(w^*, b^*)} \left[\frac{\|w\|}{2} + \sum_{i=1}^n c_i \right]$$

$$\max f(n) \Leftrightarrow \min \frac{1}{f(n)}$$

$$\left(\min \frac{2}{\|w\|} \right) \Leftrightarrow \min \frac{\|w\|}{2}$$

$C_i = 0$ (for correctly classified points)
for misclassified points,



we minimize these distances
(minimize error)

SVM error

margin error + classification error

$$\text{argmin}_{(w^*, b^*)} \frac{\|w\|^2}{2} + C \sum_{i=1}^n C_i$$

hyperparameters

high loss

if C is $\uparrow$, reduce loss error instead of margin error

C is $\downarrow$, vice versa.

$(C \propto 1/\lambda)$ - margin error

kernel trick: used for non linear data.

